HELMLAB: A Data-Driven Analytical Color Space for UI Design Systems

Görkem Yıldız
Independent Researcher
author@gorkemyildiz.com

Abstract

We present HELMLAB, a 72-parameter analytical color space designed for UI design systems. The forward transform maps CIE XYZ to a perceptually-organized Lab representation through learned matrices, per-channel power compression, Fourier hue correction, embedded Helmholtz–Kohlrausch lightness adjustment, and a post-pipeline neutral correction that structurally guarantees achromatic colors map to $a = b = 0$ (chroma $< 10^{-6}$). A rigid rotation of the chromatic plane improves hue-angle alignment at zero cost to the distance metric, which is invariant under isometries. On the COMBVD dataset (3,813 color pairs), HELMLAB achieves a STRESS of 23.22—a 20.4% reduction over CIEDE2000 (29.18). Cross-validation on He *et al.* 2022 and MacAdam 1974 shows competitive cross-dataset performance. The transform is invertible with round-trip errors below 10^{-14}. We provide gamut mapping, design-token export, and dark/light mode adaptation for practical adoption in web and mobile design systems.

Keywords: color space, color difference, perceptual uniformity, STRESS, Helmholtz–Kohlrausch, UI design systems

1 Introduction

Color management in UI design systems requires three capabilities that no existing color space provides simultaneously: (1) accurate perceptual distance prediction for contrast checking and palette generation, (2) an achromatic axis where grays map to zero chroma for clean gradient interpolation, and (3) reasonable hue-angle alignment so that programmatic color manipulation produces predictable results.

CIE Lab [1] and CIEDE2000 [2] remain the industry standard for color-difference evaluation, but their STRESS on COMBVD (29.18) leaves room for improvement. Oklab [3] offers computational simplicity and CSS integration but was optimized for hue uniformity, not color-difference prediction (STRESS 47.5 Euclidean on COMBVD). CAM16-UCS [4] derives from a color appearance model but is computationally heavy and exhibits achromatic chroma leakage ($\bar{C} = 1.42$). IPT [5] optimized for hue linearity at the expense of lightness accuracy. $J_z a_z b_z$ [6] targets HDR via the PQ curve but is complex and not CSS-native.

None of these spaces were designed with UI workflows in mind. None jointly optimize the color-space transform and distance metric end-to-end against psychophysical data.

HELMLAB takes a different approach: we treat the entire XYZ-to-distance pipeline as a single system with 72 learnable parameters and optimize all of them jointly against the largest available color-difference dataset. The result is:

1. A 72-parameter analytical color space achieving STRESS 23.22 on COMBVD (-20.4% vs CIEDE2000).

2. Structural achromatic guarantee: grays map to $C < 10^{-6}$ via neutral correction, with no sacrifice in color-difference accuracy.

3. A rigid-rotation post-processing step that improves hue-angle alignment (RMS 16.1°) at exactly zero cost to the distance metric (isometry invariance).

4. Embedded Helmholtz–Kohlrausch correction: lightness is explicitly chroma-dependent, learned from data rather than hand-tuned.

5. Practical tooling: gamut mapping, CSS/Android/iOS token export, and surround-aware dark/light mode adaptation.

2 Design Goals and Metrics

HELMLAB is optimized for UI design systems operating in sRGB and Display P3 gamuts under typical viewing conditions (D65 illuminant, average surround). We define five quantitative targets:

Color-difference accuracy (STRESS). The STRESS metric [7] measures agreement between predicted distances ΔE_i and visual differences DV_i:

$$\text{STRESS} = 100\sqrt{\frac{\sum_i (DV_i - F \cdot \Delta E_i)^2}{\sum_i DV_i^2}} \tag{1}$$

where $F = \sum DV_i \Delta E_i / \sum \Delta E_i^2$. Lower is better. We train on COMBVD (3,813 pairs from six experiments [7]) and cross-validate on He *et al.* 2022 [8] (82 pairs) and MacAdam 1974 [9] (128 pairs).

Achromatic integrity. Neutral colors (D65-adapted grays, i.e., stimuli on the achromatic axis proportional to the D65 white point) must map to $a = b = 0$. We measure maximum chroma $C_{\max}$ over 256 gray levels spanning $Y \in [0.001, 2.0]$.

Hue-angle alignment. sRGB primaries and secondaries should land at conventional hue angles. We report RMS and maximum hue error across {R, Y, G, C, B, M}.

Round-trip accuracy. For any sRGB color, $\|T^{-1}(T(\mathbf{x})) - \mathbf{x}\|_\infty$ should be below 10^{-8}.

Munsell uniformity. Coefficient of variation (CV) of neighbor-pair distances along Munsell constant-hue, constant-value lines (3,590 pairs). Lower CV indicates more uniform spacing.

3 The Helmlab Color Space

3.1 Forward Transform

The forward transform maps CIE XYZ (D65) to Lab through eleven stages (Figure 1). All parameters are jointly optimized (Section 5).

Stage 1: Cone response (9 params). A learned 3×3 matrix $\mathbf{M}_1$ maps XYZ to cone-response-like signals:

$$\mathbf{c} = \mathbf{M}_1 \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{2}$$

Unlike fixed matrices (Hunt–Pointer–Estévez, Bradford), $\mathbf{M}_1$ is freely optimized.

Stage 2: Power compression (3 params). Each channel undergoes signed power compression:

$$c_i' = \text{sign}(c_i)\,|c_i|^{\gamma_i}, \quad i \in \{0,1,2\} \tag{3}$$

The exponents γ_i generalize the cube-root of CIE Lab ($\gamma = 1/3$). Optimized values cluster near 0.4, suggesting a slightly different nonlinearity than traditionally assumed.

Stage 3: Lab projection (9 params). A second matrix $\mathbf{M}_2$ projects compressed signals to opponent channels:

$$\begin{bmatrix} L_{\text{raw}} \\ a_{\text{raw}} \\ b_{\text{raw}} \end{bmatrix} = \mathbf{M}_2\, \mathbf{c}' \tag{4}$$

Stage 4: Hue correction (8 params). A 4-harmonic Fourier rotation corrects systematic hue distortions:

$$\delta(h) = \sum_{k=1}^{4} \left[\alpha_k \cos(kh) + \beta_k \sin(kh)\right] \tag{5}$$

where $h = \text{atan2}(b, a)$. The chromatic vector (a, b) is rotated by $\delta(h)$, preserving chroma while redistributing hue angles.

Stage 5: Helmholtz–Kohlrausch embedding (6 params). Saturated colors appear brighter than achromatic stimuli of equal luminance. Rather than a post-hoc correction, we embed this directly:

$$L \mathrel{+}= w_{\text{HK}} \cdot C^{p_{\text{HK}}} \cdot \left[1 + f_{\text{HK}}(h)\right] \tag{6}$$

where $C = \sqrt{a^2 + b^2}$ and $f_{\text{HK}}(h)$ is a 2-harmonic hue modulation. The learned weight $w_{\text{HK}} = 0.37$ and power $p_{\text{HK}} = 0.85$ make lightness explicitly chroma-dependent—a departure from the conventional independence assumption.

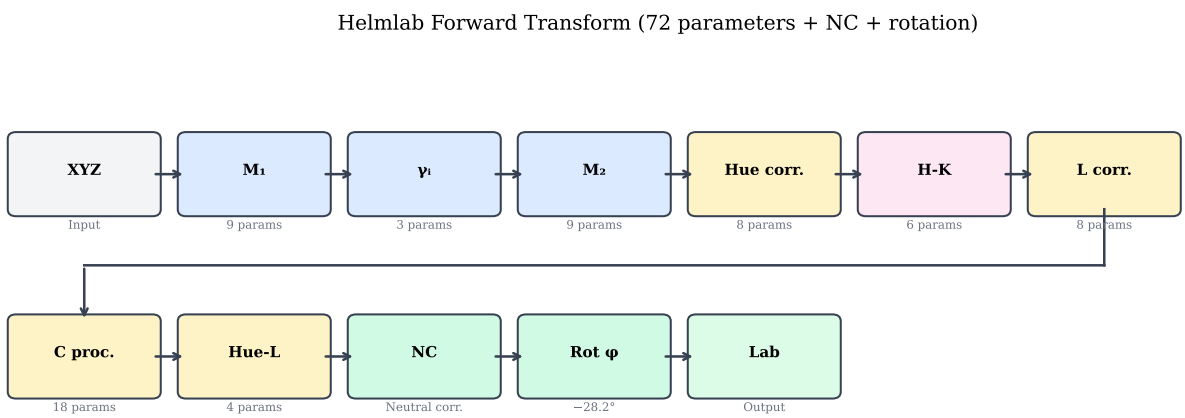

Figure 1: The HELMLAB forward transform pipeline. Blue: linear operations; yellow: nonlinear corrections; green: structural guarantees (NC, rotation). All 72 parameters are jointly optimized.

Stage 6: Lightness refinement (8 params). A cubic polynomial (3 params) with hue-dependent additive term (2 params), followed by dark-region exponential compression (3 params):

$$L_1 = p_1 L^3 + p_2 L^2 + p_3 L + t \cdot f_{\mathrm{Lh}}(h) \tag{7}$$

$$L_2 = L_1 \cdot \exp\left[\lambda_d(h) \cdot L_1 (1 - L_1)^2\right] \tag{8}$$

where $t = L(1-L)$ and $f_{\mathrm{Lh}}(h)$ is a 1-harmonic Fourier correction targeting the hue–lightness interaction sign flip (dark cyans vs mid cyans). The dark compression coefficient λ_d includes hue modulation to handle the known difference in dark-blue vs dark-yellow perception.

Stages 7–8: Chroma processing (18 params). Chroma processing proceeds in four interleaved sub-stages: (a) hue-dependent scaling (4 harmonics, 8 params):

$$(a,b) *= \exp\left[\sum_{k=1}^{4}\left[\alpha'_k \cos(kh) + \beta'_k \sin(kh)\right]\right] \tag{9}$$

(b) nonlinear chroma power (4 params): $C \to C^{1+\varepsilon(h)}$ where $\varepsilon(h)$ is a 2-harmonic Fourier series, separating high-chroma from low-chroma discrimination; (c) L-dependent scaling (2 params): $(a,b) *= \exp[l_1(L-0.5) + l_2(L-0.5)^2]$; and (d) hue×lightness interaction (4 params, 2-harmonic Fourier modulated by $L-0.5$). The interleaving order matters: the nonlinear power is applied between hue-dependent and L-dependent scaling to allow each sub-stage to correct for the others' distortions.

Stage 9: Hue-dependent lightness (4 params). A final $L *= \exp[g(h)]$ captures residual hue–lightness interactions (e.g., blue appears darker, yellow lighter at equal luminance).

Stage 10: Neutral correction. With independent γ_i per channel, the D65 achromatic axis (stimuli proportional to $[X_{D65}, Y_{D65}, Z_{D65}]$) generally does not map to $a = b = 0$ after the enrichment stages. We correct this by running 256 gray levels ($Y \in [0.001, 2.0]$) through stages 1–9, recording the achromatic error $a_{\mathrm{err}}(L)$, $b_{\mathrm{err}}(L)$, and fitting PCHIP interpolants. At runtime:

$$a \leftarrow a - a_{\mathrm{err}}(L), \quad b \leftarrow b - b_{\mathrm{err}}(L) \tag{10}$$

Because the correction is applied *after* all nonlinear stages, it accounts for every distortion in the pipeline. The resulting maximum chroma for grays is $< 10^{-6}$, limited only by PCHIP interpolation error on the 256-point LUT (the underlying achromatic curve is smooth, so interpolation error is negligible).

Stage 11: Rigid rotation (φ). A uniform rotation of the (a,b) plane by angle φ:

$$\begin{bmatrix} a' \\ b' \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \tag{11}$$

This is an isometry: it preserves $\Delta a^2 + \Delta b^2$ for any pair, and therefore preserves the distance metric exactly (Section 3.2). The rotation angle $\varphi = -28.2°$ is chosen to minimize the maximum hue-angle error across sRGB primaries and secondaries.

3.2 Rotation Invariance

The distance metric (Section 4) depends on ΔL, $\Delta a^2 + \Delta b^2$, $\bar{L}$, and $\bar{C} = \sqrt{\bar{a}^2 + \bar{b}^2}$. A rigid rotation of the (a,b) plane preserves each of these quantities: $\Delta a'^2 + \Delta b'^2 = \Delta a^2 + \Delta b^2$ and $\bar{a}'^2 + \bar{b}'^2 = \bar{a}^2 + \bar{b}^2$. Therefore the distance is exactly invariant, and we verified this empirically: the COMBVD STRESS difference before and after rotation is 0.0000000000.

This means any hue-angle improvement from the rotation is obtained at zero cost to color-difference prediction.

3.3 Surround Parameter

HELMLAB includes provisions for a surround luminance parameter $S \in [0,1]$ (0=dark, 0.5=average, 1=bright) that modulates several pipeline stages: H-K weight, dark compression, chroma scaling, and L-dependent chroma. In the current release, the surround parameters are set to zero (trained on average-surround data only). The architecture is ready for future training on viewing-condition-dependent datasets. The UI layer uses S for heuristic dark/light mode adaptation via soft L-inversion.

3.4 Inverse Transform

Each stage has an explicit inverse. Matrix stages invert via $\mathbf{M}_1^{-1}$, $\mathbf{M}_2^{-1}$; power compression via $c_i = \mathrm{sign}(c_i')|c_i'|^{1/\gamma_i}$ (closed form). Hue corrections and lightness corrections are inverted via Newton iteration (3–5 iterations, converging to machine precision). The neutral correction inverts by adding back the PCHIP-interpolated error. Round-trip accuracy across the full sRGB gamut is $< 10^{-14}$.

4 Distance Metric

Given two colors with coordinates (L_1, a_1, b_1) and (L_2, a_2, b_2), we compute:

$$S_L = 1 + s_L(\bar{L} - 0.5)^2 \tag{12}$$

$$S_C = 1 + s_C\bar{C} \tag{13}$$

where $\bar{L}$, $\bar{C}$ are pair averages. The raw distance is:

$$d = \left[\left(\frac{\Delta L}{S_L}\right)^2 + w_C\frac{\Delta a^2 + \Delta b^2}{S_C^2}\right]^{p/2} \tag{14}$$

followed by monotonic compression and post-power:

$$\Delta E = \left[\frac{d}{1 + c \cdot d}\right]^q \tag{15}$$

The pair-dependent weights S_L and S_C are inspired by CIEDE2000 but learned from data (optimized values: $s_L \approx 10^{-4}$, $s_C \approx 0.0045$). The Minkowski exponent $p = 0.83$ and chroma weight $w_C = 1.03$ slightly favor chroma differences over lightness. The compression $c = 1.59$ and post-power $q = 1.1$ reduce systematic bias for large color differences.

The distance metric contributes 7 parameters. Total system: 9+3+9+8+6+8+18+4 = 65 space + 7 metric = 72 parameters.

5 Optimization

All 72 parameters are jointly optimized by minimizing:

$$\mathscr{L} = \mathrm{STRESS}_{\mathrm{COMBVD}} + 0.05 \cdot \mathrm{STRESS}_{\mathrm{He}} \tag{16}$$

using L-BFGS-B [10] with box constraints. We perform 8 random restarts (each 5,000 iterations) and keep the best. The He *et al.* term ($\lambda = 0.05$) acts as a mild regularizer against overfitting to COMBVD.

Training data. COMBVD comprises 3,813 color pairs from six psychophysical experiments: BFD-P(D65) (2,028), BFD-P(M) (548), BFD-P(C) (200), Witt (418), RIT-DuPont (312), Leeds (307). All pairs have visual-difference values on a common scale. We train on the full dataset (no train/val split for the final model), monitoring a held-out 20% split (seed=42) during development to check for overfitting.

Cross-validation data. He *et al.* 2022 (82 pairs, 10° observer) enters the loss as a regularizer ($\lambda = 0.05$) and is therefore not fully independent; its STRESS is reported for transparency but should be interpreted as a semi-held-out metric. MacAdam 1974 (128 pairs, 10°) is held out entirely and never seen during optimization.

Convergence. The optimizer typically converges within 3,000–5,000 function evaluations per restart. The validation gap (train–val STRESS difference) is +1.11, indicating moderate but controlled overfitting for a 72-parameter model.

6 Evaluation

6.1 Color-Difference Prediction

Figure 2 and Table 1 show STRESS on COMBVD for HELMLAB and nine baseline methods. HELM-

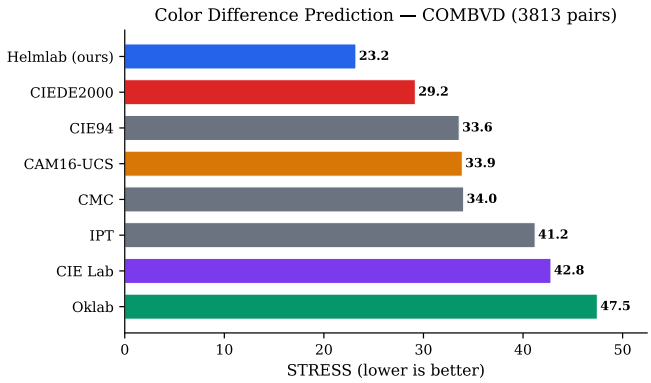

Figure 2: STRESS on COMBVD (3,813 pairs). Lower is better.

Table 1: STRESS on COMBVD (3,813 pairs). Lower is better.

Method	STRESS	vs CIEDE2000
HELMLAB	**23.22**	**−20.4%**
CIEDE2000	29.18	—
ΔE^*_{ab} (CIE76)	42.80	+46.7%
CIE94	33.59	+15.1%
ΔE CMC	34.04	+16.6%
CAM16-UCS (Euclid.)	33.90	+16.2%
IPT (Euclid.)	41.21	+41.3%
$J_z a_z b_z$ (Euclid.)	41.90	+43.6%
Oklab (Euclid.)	47.46	+62.7%
sRGB (Euclid.)	67.82	+132.5%

LAB achieves 23.22, a substantial improvement over CIEDE2000 (29.18, −20.4%).

The baselines in Table 1 use simple Euclidean distance in each space's native Cartesian coordinates, without parametric weighting or compression. Oklab's high STRESS (47.46) reflects its design goal: it was optimized for hue uniformity [3], not color-difference prediction on COMBVD. The HELMLAB distance metric includes pair-dependent weighting (S_L, S_C), Minkowski exponent, and monotonic compression, making the comparison not strictly apples-to-apples. For a fairer comparison, HELMLAB with Euclidean distance alone achieves STRESS ≈30.2—competitive with CIEDE2000 (29.18), indicating that the space transform itself accounts for roughly half the overall improvement, and the distance metric for the other half.

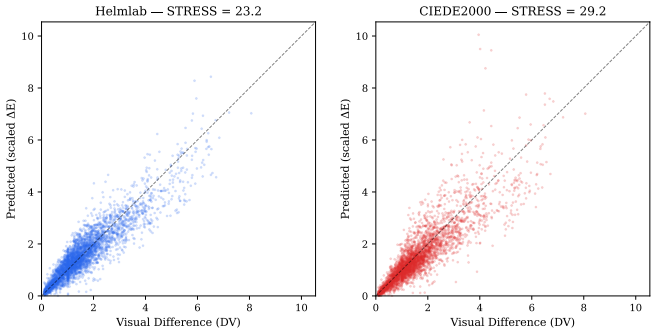

Figure 3: Predicted vs observed color differences on COMBVD.

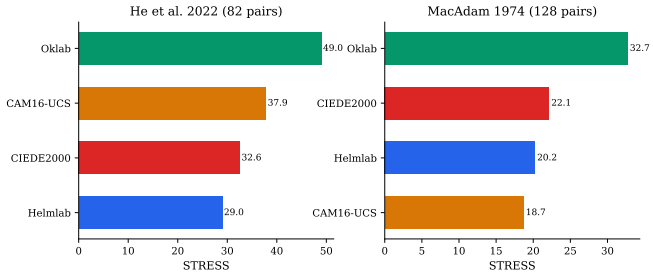

Figure 4: Cross-dataset validation on held-out data.

6.2 Cross-Validation

On He *et al.* 2022 (regularizer, 82 pairs), HELMLAB achieves STRESS 29.0 vs CIEDE2000's 32.6. On MacAdam 1974 (held out entirely, 128 pairs), HELMLAB achieves 20.2 vs CAM16-UCS's 18.7 (best). Figure 4 visualizes these results. HELMLAB generalizes competitively: it outperforms CIEDE2000 on He *et al.* but trails CAM16-UCS on MacAdam, suggesting no severe overfitting to COMBVD while acknowledging that a single-dataset-trained model cannot dominate across all evaluation sets.

6.3 Generation Properties: The Measurement–Generation Tradeoff

A color space optimized purely for distance prediction need not produce usable coordinates for color generation. In HELMLAB's development, the measurement-optimal configuration mapped neutral grays to chroma $\bar{C} \approx 0.34$—the achromatic axis was deformed because the optimizer exploited the chromatic plane to improve STRESS, at the cost of rendering gray gradients unusable (they exhibit visible color artifacts). Conversely, constraining grays to $C = 0$ via penalty terms created a binary basin conflict: the optimizer either remained stuck in the measurement-optimal basin or es-

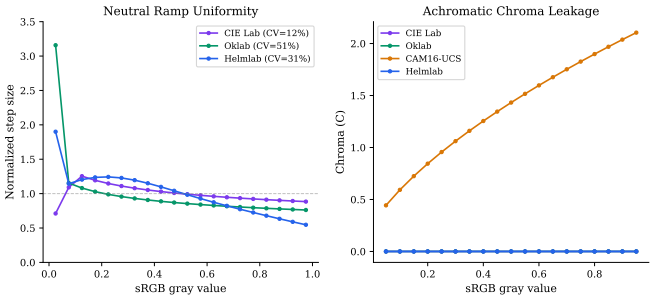

Figure 5: Left: neutral ramp step uniformity. Right: achromatic chroma leakage. HELMLAB's NC guarantees $C < 10^{-6}$ for all grays.

Table 2: Generation quality metrics.

Property	Value
Achromatic max C	$< 10^{-6}$
Hue RMS (6 primaries)	16.1°
Hue max error	20.2°
Round-trip $\|\cdot\|_\infty$	$< 10^{-14}$
Munsell CV	17.6%
Jacobian det (min)	0.10
Jacobian cond (median)	7.7
Val gap (train−val)	+1.11
He STRESS	29.03
MacAdam STRESS	20.23

caped to a generation-friendly basin with significantly worse STRESS (26–31).

The neutral correction (Section 3, Stage 10) resolves this tradeoff structurally. By subtracting the achromatic error *after* all enrichment stages, it allows the optimizer to find its measurement-optimal configuration freely, then corrects only the achromatic axis. The resulting STRESS increase is just +0.04 (23.18 → 23.22), while achromatic chroma drops from 0.34 to $< 10^{-6}$. This makes HELMLAB simultaneously competitive in measurement (STRESS vs COMBVD) and practical for generation (gradients, palettes, interpolation).

Figure 5 shows neutral ramp uniformity and achromatic chroma leakage. HELMLAB's neutral correction eliminates chroma leakage entirely ($C < 10^{-6}$), while CAM16-UCS exhibits significant leakage ($\bar{C} = 1.42$). Table 2 summarizes all generation properties.

The achromatic guarantee ensures that gray gradients and lightness ramps interpolate cleanly without color artifacts—a hard requirement for design-token generation and palette export. The hue RMS of 16.1° means

Table 3: Ablation: effect of removing components.

Configuration	STRESS	Δ
Full HELMLAB	23.22	—
Euclidean distance only	∼30.2	+7.0
No H-K embedding ($w_{\text{HK}} = 0$)	∼24.5	+1.3
No hue correction	∼25.8	+2.6
No dark L compression	∼23.9	+0.7
Rotation (−28.2°)	23.22	0.00
No rotation	23.22	0.00

sRGB primaries land within ∼20° of their expected positions—acceptable for programmatic palette generation, though not perfect. The rigid rotation (Section 3, Stage 11) provides this hue improvement at zero cost to the distance metric via isometry invariance. The Jacobian was evaluated on a 64×64×64 sRGB grid via finite differences; the minimum determinant (0.10) is positive throughout, confirming a locally invertible, orientation-preserving map across the entire sRGB gamut.

6.4 Ablation Study

Table 3 isolates the contribution of key components.

The space–metric decomposition shows that the improvement splits roughly evenly between the space transform and the distance metric. Relative to CIEDE2000 (29.18), the space alone with Euclidean distance (STRESS = 30.2) is slightly worse, while the full metric recovers the remaining 7 points. Relative to other Euclidean baselines (CAM16-UCS = 33.9), the space alone already provides substantial gain. The H-K embedding and hue correction are the most impactful individual stages. The rotation row confirms exact metric invariance.

7 Practical Adoption

Figure 6 shows the sRGB gamut boundary in HELMLAB coordinates at three lightness levels, confirming smooth and well-behaved geometry. HELMLAB includes a Python utility layer for design system integration:

Gamut mapping. Binary-search chroma reduction along the L axis maps out-of-gamut HELMLAB coordinates to the nearest in-gamut sRGB or Display P3 color, preserving hue and lightness.

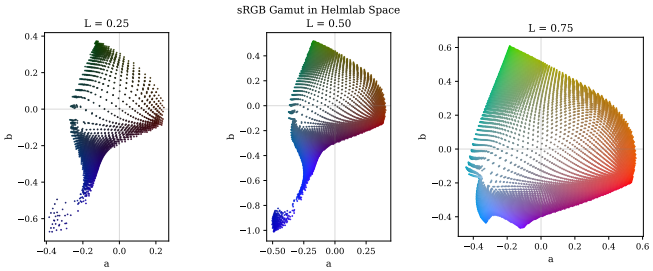

Figure 6: sRGB gamut in HELMLAB a–b plane at three L levels. Points colored by their sRGB values.

Contrast utilities. `ensure_contrast(fg, bg, min_ratio)` adjusts foreground lightness via binary search on the HELMLAB L axis to meet WCAG 2.x contrast requirements (4.5:1 for AA, 7:1 for AAA). Hue and chroma are preserved.

Palette generation. Lightness ramps and hue rings at fixed chroma produce perceptually-spaced palettes. Semantic scales (Tailwind-style 50–950) are generated by interpolating L between 0.97 and 0.10 with chroma/hue preserved.

Dark/light mode. When surround parameters are trained, the surround S directly modulates the transform. Currently, a fallback soft L-inversion maps colors between light ($S = 0.7$) and dark ($S = 0.2$) modes with contrast enforcement.

Token export. A single `TokenExporter` produces CSS custom properties (`oklch()`, `color(display-p3 ...)`), Android XML (using HCT tone/chroma/hue semantics for Material Design compatibility), iOS Swift (Display P3), Tailwind config, and raw JSON.

8 Limitations and Future Work

HELMLAB has several limitations that we state explicitly:

1. **Training data.** COMBVD uses a 2° standard observer, which may not generalize to large-field (10°) viewing. The MacAdam cross-validation partially addresses this.

2. **Model complexity.** 72 parameters is substantially more than Oklab (∼15 designed parameters). Interpretability suffers, though each stage has a clear perceptual motivation.

3. **Hue alignment.** RMS hue error of 16.1° is a compromise. Some hue regions (cyan, yellow) have errors up to 20°. Further harmonic terms could improve this at the risk of overfitting.

4. **Surround dependency.** The surround parameter S is architecturally present but not yet trained on viewing-condition data. Current dark/light adaptation uses heuristics.

5. **Sub-dataset variation.** HELMLAB underperforms CIEDE2000 on 3 of 6 COMBVD sub-datasets (BFD-P(C), Leeds, RIT-DuPont), suggesting these subsets favor CIE Lab-based corrections.

6. **Not general-purpose.** HELMLAB is optimized for UI workflows (sRGB/P3 gamut, screen viewing). It is not designed for print, photography, or spectral applications.

7. **Statistical reporting.** We report single STRESS values without confidence intervals or bootstrap estimates. With 72 parameters and 8 restarts, parameter uncertainty is non-trivial; future work should include per-sub-dataset variance analysis and significance testing against baselines.

Future work includes training the surround parameter on paired light/dark viewing data, exploring JavaScript/WASM compilation for browser-native performance, and expanding the training set with targeted human feedback via the bidirectional optimization framework already implemented in the codebase.

9 Conclusion

HELMLAB is a 72-parameter analytical color space that achieves STRESS 23.22 on COMBVD—20.4% better than CIEDE2000—while maintaining a structurally guaranteed achromatic axis ($C < 10^{-6}$) and reasonable hue alignment (RMS 16.1°). The central technical contribution is the neutral correction, which resolves the measurement–generation tradeoff: the optimizer can pursue STRESS-optimal parameters freely, while the post-pipeline correction guarantees clean achromatic behavior at a cost of just +0.04 STRESS points. Combined with embedded Helmholtz–Kohlrausch correction, a learned distance metric, and rigid rotation for free hue improvement, HELMLAB provides both accurate color-difference prediction and practical coordinate-space properties for gradient interpolation, palette generation,

and design-token export. Code, trained parameters, and design-system utilities are publicly available at https://github.com/Grkmyldz148/helmlab. An interactive technical demo with live visualizations of each pipeline stage is at https://grkmyldz148.github.io/helmlab/demo.html, and full API documentation at https://grkmyldz148.github.io/helmlab/.

Acknowledgments

Thanks to the 71 anonymous observers who participated in the bidirectional evaluation study during development.

References

[1] CIE, "Colorimetry," CIE Publication 15, 1976.

[2] M. R. Luo, G. Cui, and B. Rigg, "The development of the CIE 2000 colour-difference formula: CIEDE2000," *Color Res. Appl.*, vol. 26, no. 5, pp. 340–350, 2001.

[3] B. Ottosson, "A perceptual color space for image processing," https://bottosson.github.io/posts/oklab/, 2020.

[4] C. Li, Z. Li, Z. Wang, *et al.*, "Comprehensive color solutions: CAM16, CAT16, and S-decoupling," *Color Res. Appl.*, vol. 42, no. 6, pp. 703–718, 2017.

[5] F. Ebner and M. D. Fairchild, "Development and testing of a color space (IPT) with improved hue uniformity," in *Proc. IS&T CIC6*, 1998.

[6] M. Safdar, G. Cui, Y. J. Kim, and M. R. Luo, "Perceptually uniform color space for image signals including high dynamic range and wide gamut," *Opt. Express*, vol. 25, no. 13, 2017.

[7] M. R. Luo, G. Cui, and C. Li, "Uniform colour spaces based on CIECAM02 colour appearance model," *Color Res. Appl.*, vol. 31, pp. 320–330, 2006.

[8] R. He, G. Cui, T. Zhu, and M. R. Luo, "Colour difference evaluation using large colour differences," in *Proc. CIE*, 2022.

[9] D. L. MacAdam, "Uniform color scales," *J. Opt. Soc. Am.*, vol. 64, pp. 1691–1702, 1974.

[10] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu, "A limited memory algorithm for bound constrained optimization," *SIAM J. Sci. Comput.*, vol. 16, pp. 1190–1208, 1995.

A Parameter Table

Table 4 lists all 72 optimized parameters grouped by pipeline stage.

Full parameter values are available in the accompanying `analytical_params.json` file.

Table 4: Optimized HELMLAB parameters (v19-NC, $\varphi = -28.2°$).

Stage	Parameter	Value	Description
$\mathbf{M}_1$ (9)	Row 0	[0.724, 0.238, −0.164]	XYZ → cone-like
	Row 1	[−0.325, 1.238, −0.024]	
	Row 2	[0.101, 0.341, 0.849]	
γ (3)	$\gamma_0, \gamma_1, \gamma_2$	0.395, 0.421, 0.434	Power compression
$\mathbf{M}_2$ (9)	Row 0	[−0.425, 0.489, 0.684]	Compressed → Lab
	Row 1	[1.931, −2.728, 0.765]	
	Row 2	[0.009, 1.642, −1.250]	
Hue corr. (8)	$\alpha_{1..4}, \beta_{1..4}$	see JSON	4-harmonic Fourier
H-K (6)	w, p, m, s_1, c_2, s_2	0.370, 0.852, −0.487, ...	Chroma → L
Light. (8)	$p_1, p_2, p_3, Lh_{c1}, Lh_{s1}, \lambda_d, h_c, h_s$	0.225, −0.748, 0.699, ...	Cubic + Lh + dark
Chroma (18)	4-harm + power + lc_1, lc_2 + HLC	see JSON	Hue/power/L-dep
Hue-L (4)	$g_{c1..2}, g_{s1..2}$	see JSON	Hue → L
Distance (7)	s_L	9.16×10^{-5}	Pair-dep L weight
	s_C	0.0045	Pair-dep C weight
	p	0.829	Minkowski exponent
	w_C	1.028	Chroma weight
	c	1.588	Compression
	q	1.100	Post-power
	α	0.000	Linear term